



ROI Training

Anthropic Models on Amazon Bedrock

Module 1:

Generative AI and Claude
Models on AWS

Course Contents

► **Module 1: Foundations: Generative AI and Claude Models on AWS**

Module 2: Claude API Fundamentals and Prompt Engineering

Module 3: Data Preparation and Knowledge Bases for RAG

Module 4: Tool Use and Agentic Workflows with Claude

Module 5: Security, Governance, and Guardrails

Module 6: Practical Architecture and Deployment Patterns

Module 7: Claude Code: Building Bedrock Applications

Module 8: Amazon Kiro and Enterprise Code Generation



Module Objectives



Claude Model Family

Identify Opus, Sonnet, and Haiku use cases based on intelligence, speed, and cost



Amazon Bedrock Console

Access Claude models and enable model access in your AWS account



Cost Optimization

Balance region selection, latency, model choice, and quotas for cost efficiency



Value Propositions

Explain enterprise benefits: managed infrastructure, security, AWS integration

What Makes Generative AI Different?

- Traditional ML: Classification from fixed categories
- Generative AI: Creates new content from patterns

“The key shift: from ‘Which box?’ to ‘What should I create?’”

Why LLMs Matter for Enterprise

Unstructured Data

Documents, emails,
tickets, transcripts

Translation

Business requests to
technical specifications

Natural Language

Understanding meaning
at enterprise scale

Workflow Power

Augment processes
that needed humans

Enterprise AI Requirements

1

Control — Content filtering, topic restrictions

2

Integration — Proprietary data, APIs, systems

3

Security — Encryption, IAM, audit trails

4

Operations — Cost tracking, latency, quotas



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The Claude Model Family

The Claude Model Family

Opus 4.6

Most intelligent
For complex tasks
\$5/\$25 per MTok

Sonnet 4.6

Best balance of
speed and intelligence
\$3/\$15 per MTok

Haiku 4.5

Fastest model for
high-volume tasks
\$1/\$5 per MTok

Claude Model Specifications

Opus 4.6

\$5/\$25 per MTok

1M token context

128k max output

Highest intelligence

Sonnet 4.6

\$3/\$15 per MTok

1M token context

64k max output

Balanced default

Haiku 4.5

\$1/\$5 per MTok

200k token context

64k max output

Fastest, cheapest

When to Use Each Model

Opus

Legal analysis
Research synthesis
Agent planning
Complex reasoning

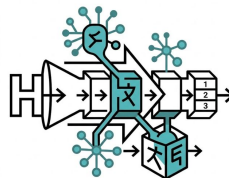
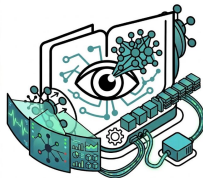
Sonnet

Customer chatbots
RAG applications
Most workloads
General purpose

Haiku

Classification
Moderation
Front-door routing
High-volume tasks

Shared Claude Capabilities



Vision

Photos, screenshots,
diagrams as input

Streaming

Real-time text
output for UIs

Multilingual

Strong across
many languages

Tools and Thinking

Function calling,
chain-of-thought

Model Selection Framework

Requirement	Recommended Model
Maximum intelligence, complex reasoning	Opus 4.6
Production workloads, balanced performance	Sonnet 4.6
High volume, cost-sensitive, speed-critical	Haiku 4.5



ROI Training

Amazon Bedrock Overview

Amazon Bedrock



Fully Managed
Foundation
Model Service

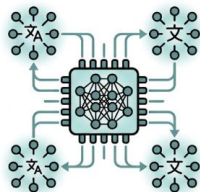
- Access Claude, Llama, Titan, and Stability AI from a single API
- Enterprise security with IAM, VPC endpoints, and CloudTrail
- No infrastructure to provision or manage—fully serverless
- Pay-per-token pricing with no upfront commitments

Amazon Bedrock Value Proposition



Fully Managed

No infrastructure to maintain



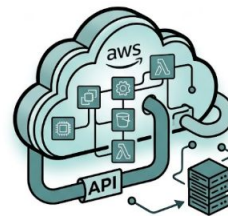
Multi-Model

Claude, Llama, Titan from one API



Enterprise-Ready

IAM, VPC, encryption built in



AWS Integration

S3, Lambda, CloudWatch native

Enterprise Security Features

1

IAM integration: Fine-grained access control

2

VPC endpoints: Keep traffic on your private network

3

Encryption: Data encrypted at rest and in transit

4

CloudTrail: Full audit logging of all API calls

5

Data isolation: Your data is never used to train models

The Amazon Bedrock Console

The screenshot displays the Amazon Bedrock console interface. The top navigation bar includes the AWS logo, a search bar with 'Bedrock' entered, and account information for 'N. Virginia' and '1234-5678-9012'. The left sidebar lists navigation options: Foundation models, Custom models, Knowledge bases, Agents, Guardrails, Playgrounds, and Model access. The main content area is titled 'Foundation Models Dashboard' and shows a summary of 18 available models, 3 custom models, and 2 knowledge bases. Three featured model cards are displayed: Anthropic Claude 3.6 Opus, Meta Llama 3 70B Instruct, and Amazon Titan Text G1 Express. Each card includes a description, a 'Text generation' tag, a 'Highly capable' tag, a 'Request access (if needed)' link, and a 'Select model' button. Below these cards is a table titled 'All available models' with columns for Provider, Model Name, Modality, Use Case, and Status. The table lists five models: Anthropic Claude 3.6 Opus, Anthropic Claude 3.6 Opus, Meta Llama 3 70B Instruct, Titan Text G1 Express, and Other Titan Models, all with a status of 'Available'. On the right side of the dashboard, there is a 'Get Started' section with two steps and links to documentation and Bedrock. Below that is a 'Recent usage' section with a line graph showing usage over time. At the bottom right is a 'Billing' section with a line graph showing billing over time.

Amazon Bedrock

Foundation Models Dashboard

18 Available Models, 3 Custom Models, 2 Knowledge Bases

Anthropic Claude 3.6 Opus

Anthropic Claude 3.6 Opus is a highly capable, multimodal context and multimodal, and thousands of other tasks.

Text generation Highly capable

Request access (if needed)

Select model Status

Meta Llama 3 70B Instruct

Llama 3 70B Instruct is a model suitable for even zero-shot for applications that require high-quality, natural language generation.

Text generation Highly capable

Select model Status

Amazon Titan Text G1 Express

Amazon Titan Text G1 Express is a model that makes it easy to integrate with your existing applications.

Text generation Highly capable

Select model Status

All available models

Provider	Model Name	Modality	Use Case	Status
Anthropic	Claude 3.6 Opus	Text	Text generation	Available
Anthropic	Claude 3.6 Opus	Text	Text generation	Available
Meta	Llama 3 70B Instruct	Text	Text generation	Available
Titan	Titan Text G1 Express	Text	Text generation	Available
Other	Titan Models	Text	Text generation	Available

Get Started

- Get the foundation models and running a operational across configuring to our models.
- Get a role

[Documentation](#)
[Bedrock](#)

Recent usage

Billing

Demo: Amazon Bedrock Playground

The screenshot displays the Amazon Bedrock Playground interface. At the top, the AWS logo and navigation bar are visible, including 'Services', a search bar, and account information for 'N. Virginia' and 'BedrockUser@1234-5678-9012'. The breadcrumb trail shows 'Amazon Bedrock > Playgrounds > Chat'.

The main section is titled 'Chat'. On the left, a model selection dropdown is highlighted with a red box. It shows 'Model: Claude v3 Sonnet' with a 'Sonnet 4.6' tag, and the full model ID 'claude-3-sonnet-20240307'. Below this, the selected model is 'Claude v3 Sonnet (anthropic.claude-v3-sonnet:0)'.

The chat area contains a prompt: 'Explain Amazon Bedrock in three sentences.' The response from the model is: 'Amazon Bedrock is a fully managed service that offers a choice of high-performing foundation models (FMs) from leading AI companies via a single API. It provides a wide range of capabilities for building and scaling generative AI applications with security and privacy, while offering tool integration and model evaluation features. By abstracting the complexity of infrastructure management, it allows organizations to experiment with and deploy various FMs efficiently and responsibly.'

Below the response is an input field with the same prompt 'Explain Amazon Bedrock in three sentences.' and a 'Send' button.

On the right side, the 'Model parameters' panel is visible, showing sliders for 'Temperature' (0.5), 'Top P' (1.0), and 'Maximum response length' (512 tokens). It also has a 'Stop sequences' dropdown set to 'None' and a 'Penalties' slider.

At the bottom of the chat area, a status bar shows 'Input tokens: 50 | Output tokens: 80 | Latency: 1.2s'.

The footer of the interface includes a 'Console' button, the region 'N. Virginia', and links for 'Dittole', 'Protry', and 'Billing account'.

Claude Model Regional Availability



AWS Americas

us-east-1, us-east-2 (Ohio)
us-west-1, us-west-2 (Oregon)
ca-central-1, sa-east-1



AWS Europe and Asia

eu-west-1, eu-west-2, eu-central-1
ap-northeast-1 (Tokyo)
ap-southeast-1/2, me-south-1

AWS GovCloud and Anthropic



Compliant Cloud
Regions for
Regulated
Workloads

Claude Sonnet 4.5 runs in us-gov-west-1 and us-gov-east-1, supporting FedRAMP High compliance. Isolated infrastructure for government workloads. Other Claude models are coming soon to GovCloud.

Region Selection Trade-offs

Latency

Closer regions deliver faster responses; cross-ocean adds 80-100ms round-trip.

Compliance

Data residency rules may force a specific region for regulated workloads.

Pricing

Token pricing is consistent, but data transfer costs vary by region.

Availability

Not all Claude models are in every region — verify before you commit.

Claude Model Pricing on Amazon Bedrock

Model	Input (per MTok)	Output (per MTok)	Best For
Claude Opus 4.6	\$5	\$25	Complex reasoning, agents
Claude Sonnet 4.6	\$3	\$15	Balanced speed/intelligence
Claude Haiku 4.5	\$1	\$5	High-volume, fast tasks

Cross-Region Inference Profiles

Global Routing

- Global: Dynamic routing for max availability
- Regional: Guaranteed routing through a specific region
- Cross-region profiles for global deployment
- Trade-off: control over data location vs. uptime
- Choose based on compliance and SLA needs



Multi-Region Deployment Pattern

1

Primary region for normal operations (regional endpoint)

2

Secondary region for failover (regional endpoint)

3

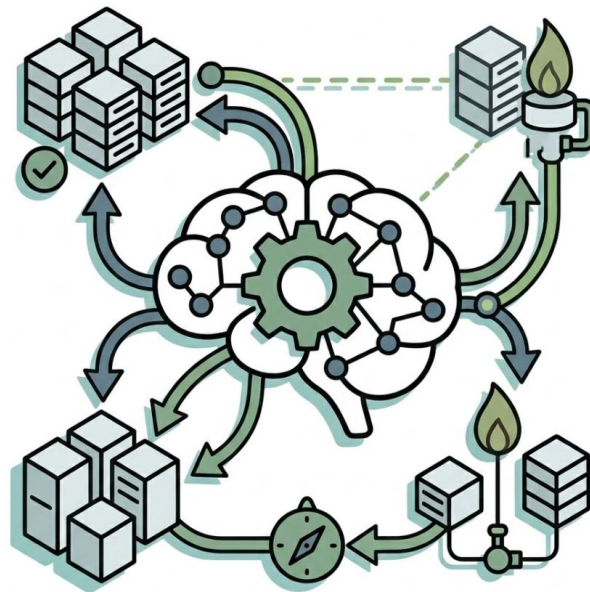
Application-level health checks and routing

4

Cross-region inference for global availability tier

Resilience Best Practices

- Implement exponential backoff with jitter for retries
- Monitor CloudWatch metrics for throttling and errors
- Test failover procedures regularly
- Consider warm standby vs. pilot light approaches



What We Covered



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Module 2 Is Next!

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